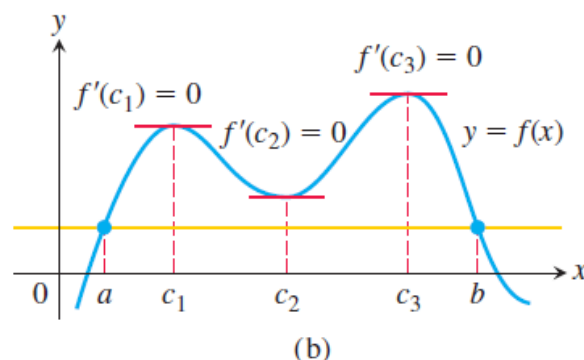
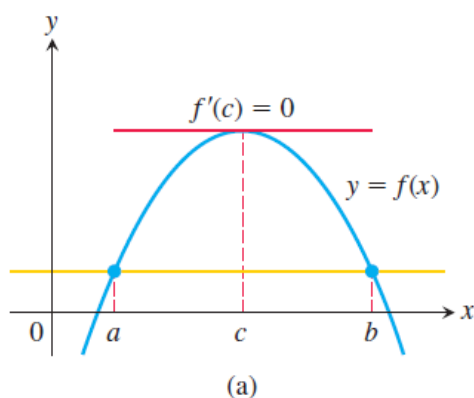


4.2 The Mean Value Theorem

We know that constant functions have zero derivatives, but could there be a more complicated function whose derivative is always zero? If two functions have identical derivatives over an interval, how are the functions related? We answer these and other questions in this section by applying the Mean Value Theorem.

① Rolle's Theorem

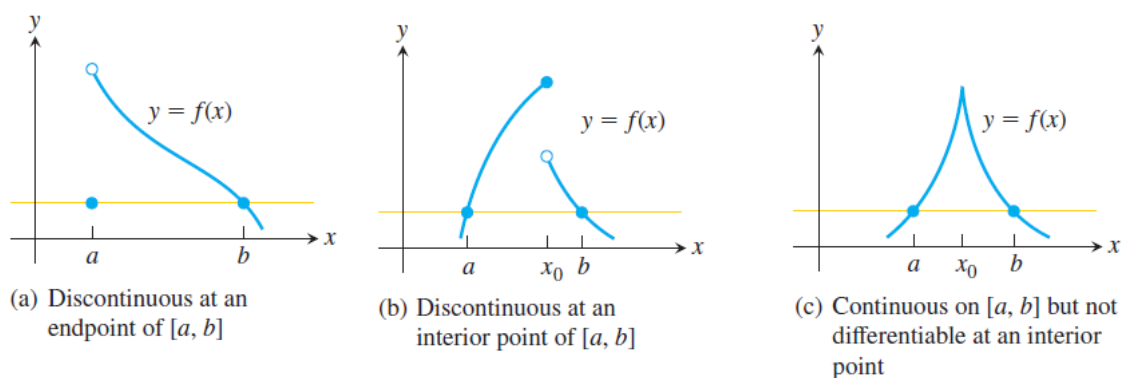
As suggested by its graph, if a differentiable function crosses a horizontal line at two different points, there is at least one point between them where the tangent to the graph is horizontal and the derivative is zero. We now state and prove this result.



THEOREM 3 (Rolle's Theorem)

Suppose that $y = f(x)$ is continuous over the closed interval $[a, b]$ and differentiable at every point of its interior (a, b) . If $f(a) = f(b)$, then there is at least one number c in (a, b) at which $f'(c) = 0$.

Note 1 The hypotheses of Theorem 3 are essential. If they fail at even one point, the graph may not have a horizontal tangent.



Example 1 Show that the equation

$$x^3 + 3x + 1 = 0$$

has exactly one real solution.

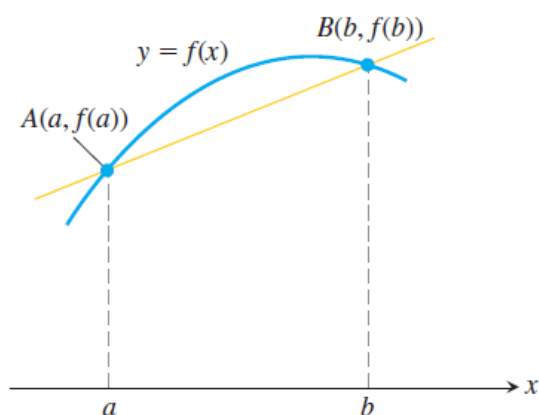
② The Mean Value Theorem

The Mean Value Theorem, which was first stated by Joseph-Louis Lagrange, is a slanted version of Rolle's Theorem. The Mean Value Theorem guarantees that there is a point where the tangent line is parallel to the secant line that joins A and B .

THEOREM 4 (The Mean Value Theorem)

Suppose $y = f(x)$ is continuous over a closed interval $[a, b]$ and differentiable on the interval's interior (a, b) . Then there is at least one point c in (a, b) at which

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$



■ A Physical Interpretation

We can think of the number

$$\frac{f(b) - f(a)}{b - a}$$

as the average change in f over $[a, b]$ and $f'(c)$ as an instantaneous change. Then the Mean Value Theorem says that the instantaneous change at some interior point is equal to the average change over the entire interval.

At the beginning of the section, we asked what kind of function has a zero derivative over an interval. The first corollary of the Mean Value Theorem provides the answer that only constant functions have zero derivatives.

Note 2 If $f'(x) = 0$ on (a, b) , then

$$f(x) = C$$

for all $x \in (a, b)$, where C is a constant.

Note 3 If $f'(x) = g'(x)$ on (a, b) , then there exists a constant C such that

$$f(x) = g(x) + C$$

for all $x \in (a, b)$. That is, $f - g$ is a constant function on (a, b) .

Example 2 Find the function $f(x)$ whose derivative is $\sin x$ and whose graph passes through the point $(0, 2)$.